A Semi- Supervised Active Learning Neural Network for Data Streams With Concept Drift

A Project Report submitted in partial fulfillment of the requirements for the award of the degree of

BACHELORS OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

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2025

DECLARATION

I hereby declare that the project report entitled "**A Semi- Supervised Active Learning Neural Network for Data Streams With Concept Drift**" is an original work done in the Department of Computer Science and Engineering /Department of Artificial Intelligence and Data Science, GITAM School of CSE, GITAM (Deemed to be University) submitted in partial fulfillment of the requirements for the award of the degree of B.Tech. in Computer Science and Engineering. The work has not been submitted to any other college or University for the award of any degree or diploma.

Date: 24th November 2025

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CERTIFICATE

This is to certify that the project report entitled "**A Semi- Supervised Active Learning Neural Network for Data Streams With Concept Drift**" is a bonafide record of work carried out by Sreeja Sistla (VU22CSEN0100301), Sanjana Sripati(VU22CSEN0100563), Lakshmi Nikhita (VU22CSEN0100440), Arun Kumar Remella(VU22CSEN0102106) students submitted in partial fulfillment of requirement for the award of degree of Bachelors of Technology in Computer Science and Engineering .

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ABSTRACT

This project addresses the challenge of building accurate machine-learning models in situations where large quantities of unlabeled streaming data are available, but manual labeling is expensive and limited. We propose a method called Dual-Teacher RBF Active Learning (DT-RBF-AL), which integrates semi-supervised learning and active learning in a closed-loop form. The method uses an RBF network as the main learner and combines two teachers—K-Nearest Neighbors (local teacher) and a secondary RBF teacher (global teacher)—to compute disagreement scores, enabling the system to select only the most informative samples for labeling. The model updates continuously as data streams in, improving with minimal human labeling. Experiments conducted on the Electricity dataset demonstrate improved accuracy, stability, and label-efficiency compared to traditional single-teacher active learning approaches.

. INTRODUCTION

Modern real-world applications such as energy demand forecasting, sensor analytics, and financial time-series modeling often involve continuous streams of data. While unlabeled data is abundant, human labeling is costly and slow. This gap has motivated research in semi-supervised learning (SSL) and active learning (AL). SSL leverages unlabeled data to enhance model performance, whereas AL intelligently selects only the most useful samples for labeling.

Existing systems often rely on a single teacher or pseudo-labeler, which limits their robustness—especially under concept drift, noise, or fluctuating input patterns. To address these limitations, this project introduces a Dual-Teacher RBF Active Learning framework that employs two complementary teachers to produce more reliable disagreement scores for active querying. The aim is to design a model that adapts continuously, maintains stability over time, and reduces labeling cost while improving accuracy.

. LITREATURE REVIEW

Past work in active learning has focused on uncertainty sampling, query-by-committee, and margin-based selection. Semi-supervised methods such as consistency regularization and pseudo-labeling have shown success in leveraging unlabeled data. The SSALD (Semi-Supervised Active Learning with Disagreement) framework introduced the idea of smoothness loss between model predictions and a teacher such as KNN. However, these systems rely on a single teacher model, limiting their robustness.

RBF networks have been studied for streaming data due to their interpretability and fast adaptation capabilities. Recent research also highlights the importance of combining local and global predictive structures for improved stability. However, no existing method combines dual teachers with RBF in a closed-loop active learning setting. This project builds upon and addresses the gaps in these works.

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PROBLEM IDENTIFICATION AND OBJECTIVES

Problem Identification

Single-teacher AL systems fail when the teacher provides noisy or unstable pseudo-labels. This leads to incorrect sampling, poor adaptation, or unnecessary label requests. The challenge is to design a system that can measure disagreement more reliably and adapt to streaming data effectively.

Objectives

- To design a dual-teacher active learning model combining local and global pseudo-labelers.
- To integrate SSL and AL into a closed-loop RBF framework for streaming data.
- To minimize human labeling while maximizing model accuracy over time.
- To visualize internal model behavior such as RBF centers and adaptation.

EXISTING SYSTEM

The SSALD approach uses a single teacher (KNN) to compute smoothness loss. While effective in simple scenarios, its performance degrades when the teacher's predictions are unstable or noisy. Moreover, single-teacher systems cannot capture both local and global structures simultaneously. As a result, they suffer from limited robustness to concept drift and may waste labeling budget on uninformative samples.

. PROPOSED SYSTEM

The proposed DT-RBF-AL system introduces two teachers:

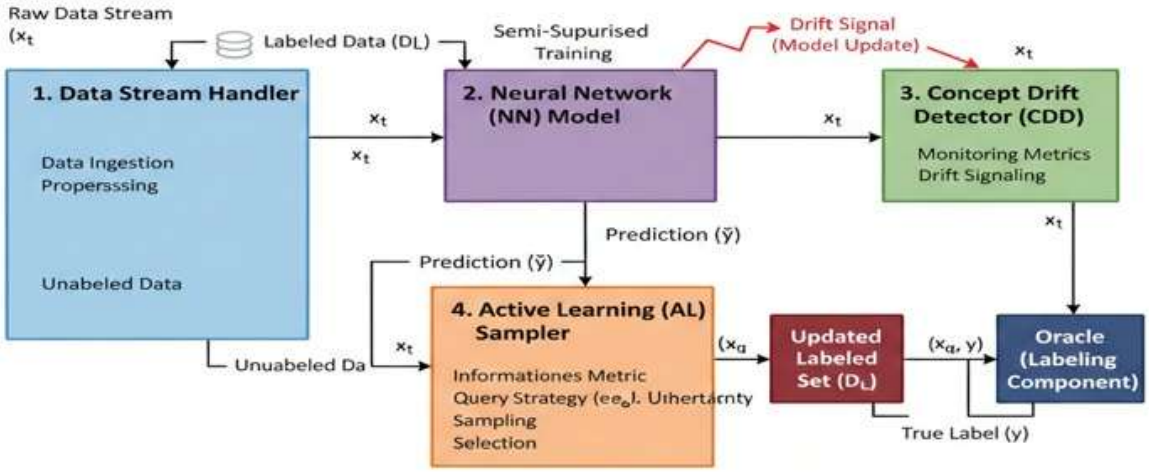
KNN Teacher (local view)

Secondary RBF Teacher (global view)

A disagreement score is computed using normalized cross-entropy from both teachers. Points where both teachers disagree with the main model receive higher scores, ensuring only informative instances are queried. The system processes data in chunks, continuously updates labeled/unlabeled pools, and incrementally trains the RBF learner. This architecture improves robustness, accuracy, and label-efficiency.

. **SYSTEM ARCHITECTURE**

1. Input streaming data
2. Preprocessing + scaling
3. Labeled pool
4. Unlabeled pool
5. RBF main learner
6. KNN teacher
7. RBF teacher
8. Disagreement computation (dual-teacher score)
9. Query selection (top-B samples)
10. Closed update



. TOOLS / TECHNOLOGIES USED

- Python
- Scikit-Learn
- NumPy / Pandas
- Matplotlib
- Jupyter Notebook
- RBF Classifier
- KNN
- Active Learning Loop Design
- StandardScaler

CONCLUSION

The project successfully implements a robust dual-teacher closed-loop SSL-AL framework for streaming data classification. DT-RBF-AL achieves higher accuracy with fewer labeled samples and shows stability under dynamic data conditions. The approach addresses a major research gap by integrating local and global consistency checks, improving reliability over single-teacher systems. The outcomes show potential for real-world deployment in any data-stream scenario where labeling cost is high.

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Table 1.1 : Results of the table

Chunk	Labeled	Test Accuracy	Train Accuracy
1	220	0.5488	0.6227
2	240	0.5488	0.6500
3	260	0.5497	0.6346
4	280	0.5479	0.6395
5	300	0.5488	0.6508
6	320	0.5534	0.6625
7	340	0.5561	0.6691
8	360	0.5588	0.6722
9	380	0.5606	0.6798
10	400	0.5652	0.6821

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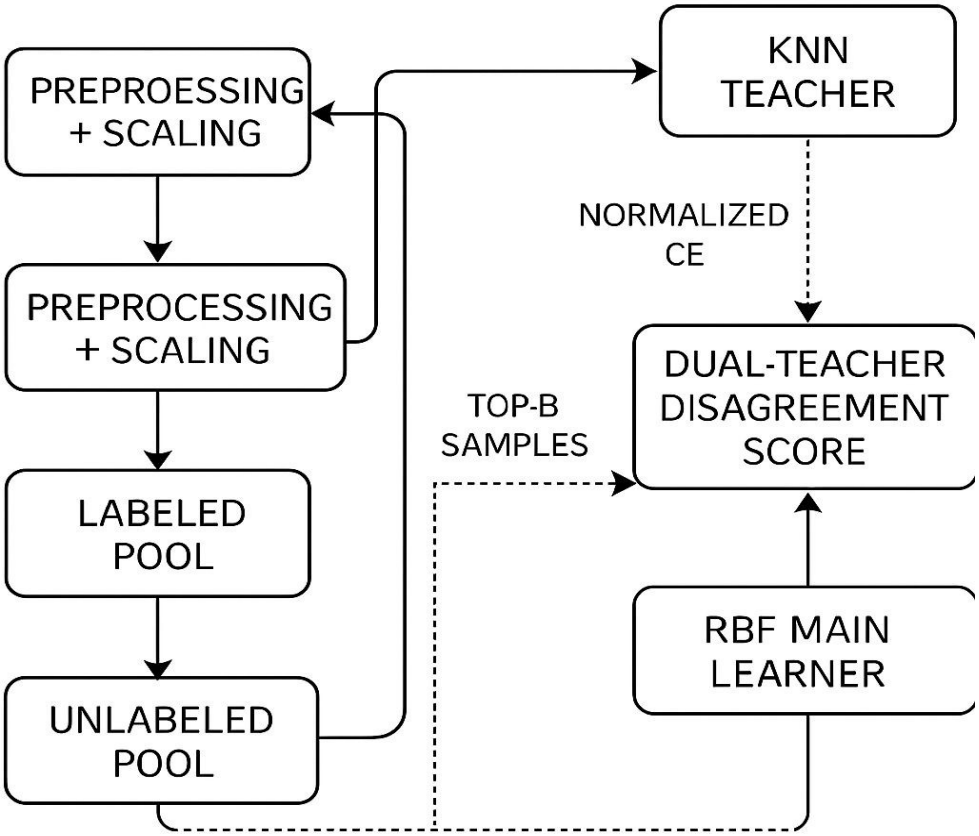
Figure 1 : Details of the Process